

LEAF & LOGIC

SMART
GREENHOUSE



Plant Research Findings for
Optimal Temperature, Soil Moisture,
LED Light Pattern & Musical Stimulation.

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Plant Research Findings for Optimal Temperature, Soil Moisture, LED Light Pattern & Musical Stimulation.

Plant Types:

Plant 1: Rubella Peperomia

Initial Height: 4"

Initial Width: 4.5"

Plant 2: Peperomia Hope

Initial Height: 5"

Initial Width: 6"

Plant 3: Peperomia Marble

Initial Height: 6.5"

Initial Width: 9"

Plant 4: Cupid Peperomia

Initial Height: 6"

Initial Width: 9"

Temperature:

Output: Reading levels are typically read in degrees.

Optimal temperature to maintain for plants: 18°C – 29°C

Specs:

- Usable with 3.0V to 5.5V power/data
- $\pm 0.5^{\circ}\text{C}$ Accuracy from -10°C to $+85^{\circ}\text{C}$
- Usable temperature range: -55 to 125°C (-67°F to $+257^{\circ}\text{F}$)
- 9 to 12 bit selectable resolution
- Uses 1-Wire interface- requires only one digital pin for communication.
- Unique 64 bit ID burned into the chip
- Multiple sensors can share one pin
- Temperature-limit alarm system

- Query time is less than 750ms.
- 3 wires interface:
 - Type A
 - Red wire – VCC
 - Black wire – GND
 - Yellow wire – DATA
 - Type B (Recently there's a series of sensors using this pin mapping. Sorry for the inconvenience.)
 - Red wire – VCC
 - Yellow wire – GND
 - Green wire – DATA
- Stainless steel tube 6mm diameter by 35mm long
- Cable diameter: 4mm (0.16)
- Length: 95cm (37.4")

Soil Moisture:

Output: Reading levels are typically read in percentages of RH (relative humidity).

Optimal soil moisture to maintain for plants: 50-70%, trigger point for irrigation should be around 40-45%.

Specs:

- Supports 3-Pin Gravity Sensor interface.
- Analog output
 - **Applications:**
 - Garden plants
 - Moisture detection
 - Intelligent agriculture
 - **Technical Specifications:**
 - **Operating Voltage:** DC 3.3-5.5V
 - **Output Voltage:** DC 0-3.0V
 - **Interface:** PH2.0-3P
 - **Size:** 99x16mm/3.9×0.63"
 - **More information:**
 - This capacitive soil moisture sensor measures soil moisture levels by capacitive sensing, rather than resistive sensing like other types of moisture sensor. It is made of corrosion-

resistant material giving it long service life.

- insert it into the soil and impress your friends with the real-time soil moisture data.
- This module includes an on-board voltage regulator which gives it an operating voltage range of 3.3 ~ 5.5V.
- This sensor is compatible with DFRobot 3-pin “Gravity” interface, which can be directly connected to the Gravity I/O expansion shield.

LED Light Patterns/Colour:

Angled lighting: Angled lighting promotes equal height and width growth for plants.

Colours: Blue, Red & White

Duration: 12 to 16 hours per day, can be indirect

Brightness Levels: moderate to light intensity 40-50%

Specs:

- **Product name:** 2 heads led grow lamp with timer
- **Input Voltage:** 100-240
- **Output Voltage:** 5V
- **LED Quantity:** 40pcs
- **Color of lamp beads:** Full Spectrum
- **Material:** Aluminum
- **Power cable:** about 2.1m
- **Frequency:** 50-60Hz
 - **More Information:**
 - **3 Lighting Modes:** warm white, red+blue, warm white+blue+red lighting modes options
 - **Timing Modes:** 6H/12H/16H automatic memory cycle lighting everyday
 - **5 Dimming Levels:** Brightness adjustable range is 20%-100%

Musical Stimulation:

Music Genre: Musical stimulation of jazz or classical music with a frequency of 115-220Hz is optimal for plant growth. Time of day to play music for plants: Morning to early afternoon.

Durations: 1-3 hours

Volume: Moderate

Things to avoid: Playing music at night, it is typically the plants resting time. Playing music too loud can cause negative effects.

Playlist chosen for plants:

https://open.spotify.com/playlist/2QqHZfdePzosdFgvHdz52h?go=1&sp_cid=e7c8ed27eafb47643f2c77b5d7c16bf0&utm_source=embed_player_p&utm_medium=desktop&nd=1&dlsi=814474ac3c544b78

Summary

Temperature:

- **Optimal Temperature:** 18-29C

Soil Moisture:

- **Optimal Soil Moisture:** 50-70%
- **Irrigation Trigger Point:** 40-45%

LED Light Pattern:

- **LED Light Colour:** Combination of Red and Blue
- **LED Timer:** 12hrs to 16 hrs
- **LED distribution:** Angled Lighting
- **LED light intensity:** 40-50%

Musical Stimulation:

- **Musical Genre:** Jazz or Classical
- **Musical Timer:** 1 to 3 hrs early afternoon
- **Musical Volume:** Moderate
- **Musical Hertz:** 115-220Hz